
Multi-label Odor Classification

Internship Progress Report

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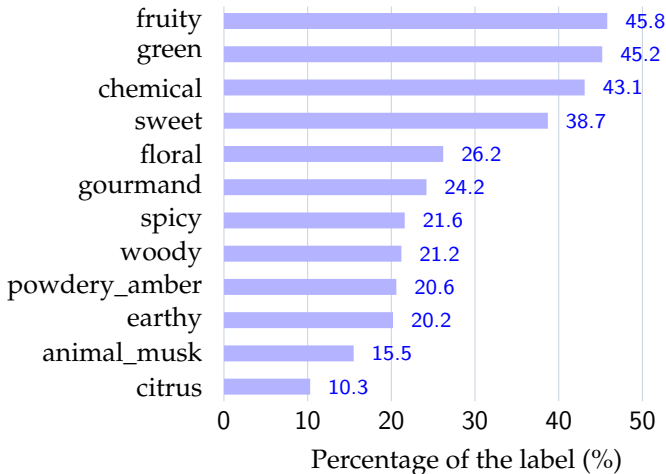
Dataset & Label Distribution

GoodScents + Leffingwell

- 4 981 molecules
- 138 raw labels → **12 metacategories**

Why reduce to 12?

- **Sparsity:** most of 138 labels have too few positives to learn from
- **Platform:** Binary Relevance trains one model per label



Molecular Feature Extraction

MACCS Keys

166 binary bits

- Presence or absence of 166 predefined substructures
- Fixed, interpretable

Morgan Fingerprints

512 bits, radius = 2

- Circular neighbourhood hashing around each atom
- Encodes local chemical environment

Mordred Descriptors

327 continuous features

- Physicochemical properties: MW, topology, logP...

Total: 1,005 features

Preprocessing Pipeline

Step 1 NaN Inspection & Row or Column removal

Exactly 5 rows have NaN across all 327 Mordred features , the same 5 rows in every column, caused by Mordred failing on unusual molecules.

Step 2 Feature Group Separation

Fingerprints and Mordred are treated separately throughout the pipeline

Step 3 Zero-Variance Removal

The columns that carry no information and are dropped via `VarianceThreshold`.

Preprocessing Pipeline

Step 4 Stratified Train/Test Split

`iterative_train_test_split` (scikit-multilearn) is used instead of standard `sklearn`, which does not handle multi-label stratification. It preserves each label's positive/negative class ratio independently in both train and test sets.

Step 5 StandardScaler on Mordred

The scaler is fitted on training data only, then applied to the test set.

$$x_{\text{scaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Step 6 Correlation Filter on Mordred

For each pair of Mordred features with $|r| > 0.95$, one feature is dropped.

The multi-label problem is decomposed into $L = 12$ independent binary classification problems. For each label $l \in \{1, \dots, 12\}$, a separate classifier f_l is trained independently. The joint distribution is factorised as:

$$P(\mathbf{y} \mid \mathbf{x}) = \prod_{l=1}^{12} P(y_l \mid \mathbf{x})$$

Assumption : label independence

Binary Relevance assumes that labels are independent given the molecule. In practice, odor labels are correlated but BR ignores this structure entirely.

Logistic Regression : Model & Loss function

Prediction For each label l , LR models:

$$P(y_l=1 \mid \mathbf{x}) = \sigma(\mathbf{w}_l^\top \mathbf{x} + b_l) = \frac{1}{1 + e^{-(\mathbf{w}_l^\top \mathbf{x} + b_l)}}$$

Weighted Loss Function

$$\mathcal{L}(\mathbf{w}_l) = - \sum_{i=1}^n w^{(i)} [y^{(i)} \log P + (1-y^{(i)}) \log(1-P)] + \frac{\|\mathbf{w}_l\|^2}{C}$$

Class Weights (class_weight='balanced')

$$w^{(i)} = \begin{cases} \frac{n}{2n_+} & y^{(i)} = 1 \\ \frac{n}{2n_-} & y^{(i)} = 0 \end{cases}$$

Solver: saga Stochastic gradient descent with variance reduction

Hyperparameter Tuning GridSearchCV

The regularisation strength C is selected by cross-validated grid search.

$$C^* = \arg \max_{C \in \mathcal{G}} \frac{1}{3} \sum_{k=1}^3 \frac{1}{12} \sum_{l=1}^{12} F1_l^{(k)}(C)$$

Search grid

$\mathcal{G} =$
 $\{0.01, 0.1, 1, 10, 100\}$

Cross-validation

3-fold CV on training data
only

Total fits

$5 \times 3 \times 12 = 180$ LR
models

Evaluation Metrics

Metric	Formula	Justification
Balanced Accuracy	$(\text{Sens.} + \text{Spec.})/2$	Corrects for class imbalance
MCC	$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Robust single score using all four confusion matrix cells
F1 Score	$2PR/(P + R)$	Optimization metric for GridSearchCV
AUC ROC	Area under ROC curve	Threshold free
AUC PR	Area under PR curve	Best for rare positive labels and robust to imbalance
Sensitivity	$TP/(TP + FN)$	Fraction of true positives caught
Specificity	$TN/(TN + FP)$	Fraction of true negatives rejected

JadBio Experiments : SMOTE vs. Original Data

Chemically invalid synthetics

SMOTE interpolates between feature vectors. For binary fingerprints, the interpolation of two bit vectors does not correspond to any real chemical structure, it introduces noise rather than signal.

JadBio handles imbalance implicitly

JadBio's aggressive feature selection discards uninformative features and concentrates on discriminative ones, partially compensating for imbalance without oversampling.

Small minority $\Rightarrow$ overfitting

For labels like animal_musk (15.5%) and citrus (10.3%), synthetic samples are more than the real . The model overfits to this region and misses the real decision boundary.

Results : BR + Logistic Regression

Label	Bal.Acc	MCC	F1	ROC AUC	PR AUC	Sens.	Spec.
floral	0.812	0.546	0.656	0.875	0.672	0.854	0.771
fruity	0.772	0.537	0.736	0.852	0.804	0.776	0.768
sweet	0.682	0.348	0.609	0.742	0.595	0.784	0.579
woody	0.771	0.454	0.564	0.837	0.567	0.763	0.779
green	0.713	0.419	0.673	0.776	0.667	0.740	0.686
spicy	0.737	0.380	0.507	0.805	0.505	0.795	0.678
animal_musk	0.703	0.302	0.406	0.797	0.442	0.662	0.743
earthy	0.738	0.395	0.515	0.820	0.571	0.700	0.776
citrus	0.766	0.354	0.393	0.851	0.390	0.735	0.797
chemical	0.717	0.423	0.668	0.779	0.654	0.765	0.670
gourmand	0.748	0.419	0.557	0.817	0.584	0.776	0.719
powdery_amber	0.739	0.393	0.515	0.810	0.465	0.727	0.751
Macro Avg	0.741	0.414	0.567	0.813	0.576	0.756	0.726

Results : BR + Random Forest

Label	Bal.Acc	MCC	F1	ROC AUC	PR AUC	Sens.	Spec.
floral	0.740	0.513	0.615	0.858	0.672	0.562	0.918
fruity	0.773	0.551	0.733	0.847	0.807	0.715	0.832
sweet	0.674	0.352	0.574	0.752	0.618	0.561	0.787
woody	0.665	0.391	0.474	0.830	0.576	0.393	0.937
green	0.718	0.434	0.668	0.790	0.713	0.676	0.760
spicy	0.673	0.422	0.494	0.818	0.585	0.400	0.947
animal_musk	0.641	0.403	0.422	0.811	0.528	0.305	0.977
earthy	0.668	0.418	0.481	0.817	0.581	0.383	0.953
citrus	0.638	0.409	0.406	0.851	0.488	0.290	0.986
chemical	0.722	0.435	0.670	0.801	0.729	0.727	0.716
gourmand	0.696	0.445	0.538	0.840	0.638	0.462	0.929
powdery_amber	0.662	0.415	0.472	0.825	0.552	0.366	0.957
Macro Avg	0.689	0.432	0.546	0.820	0.624	0.487	0.892

Results : JadBio AutoML

Label	Bal.Acc	MCC	F1	ROC AUC	PR AUC	Sens.	Spec.
floral	0.750	0.526	0.638	0.857	0.817	0.595	0.905
fruity	0.799	0.600	0.778	0.872	0.871	0.762	0.835
sweet	0.681	0.366	0.603	0.753	0.749	0.587	0.775
woody	0.691	0.468	0.530	0.845	0.800	0.427	0.955
green	0.709	0.426	0.665	0.796	0.796	0.623	0.794
spicy	0.629	0.354	0.410	0.794	0.740	0.302	0.956
animal_musk	0.665	0.449	0.472	0.805	0.753	0.354	0.975
earthy	0.683	0.470	0.516	0.802	0.776	0.401	0.964
citrus	0.606	0.370	0.334	0.842	0.741	0.221	0.991
chemical	0.728	0.466	0.678	0.800	0.787	0.639	0.817
gourmand	0.671	0.450	0.502	0.845	0.807	0.377	0.964
powdery_amber	0.628	0.317	0.403	0.792	0.717	0.323	0.933
Macro Avg	0.687	0.439	0.544	0.817	0.779	0.468	0.905

Key Observations: Three-Way Comparison

Macro Avg	BR+LR	BR+RF	JadBio
Balanced Accuracy	0.741	0.689	0.687
MCC	0.414	0.432	0.439
F1	0.567	0.546	0.544
ROC AUC	0.813	0.820	0.817
PR AUC	0.576	0.624	0.779
Sensitivity	0.756	0.487	0.468
Specificity	0.726	0.892	0.905

BR+LR

Best balanced accuracy and sensitivity .

BR+RF

Higher ROC AUC and PR AUC than LR. Better MCC.
Trades sensitivity for specificity compared to LR.

JadBio

Best PR AUC and specificity .

Improve the classifier

- BR + XGBoost / Gradient Boosting
Chacko et al. (2020):
XGBoost optimal for odor classification
- BR + SVM
- All tuned with GridSearchCV + macro-F1

Address label independence

- **Classifier Chains**

$$P(y_I \mid \mathbf{x}, y_1, \dots, y_{I-1})$$

- **Label Powerset** treat each unique label combination as a class